

A CHARACTERISTIC-FREE INTEGRAL WITNESS FOR FRÖBERG'S SERIES FOR SIX TO TWENTY-ONE SEPTICS IN FOUR VARIABLES

ABSTRACT. Fröberg's Conjecture 1, as stated in the Fröberg–Lundqvist survey, predicts the Hilbert series of a quotient of a polynomial ring by generic forms; the positive-characteristic cases the survey records are $n \leq 2$ and $n = 3$ over an infinite field. We exhibit, for every r with $6 \leq r \leq 21$, an explicit r -tuple of septics in $\mathbb{Z}[x_1, x_2, x_3, x_4]$ with all coefficients 0 or 1 whose reduction modulo the characteristic of a field k generates an ideal of $k[x_1, x_2, x_3, x_4]$ with the conjectured quotient series, for *every* field k . This is a statement about one explicit tuple and not about generic forms; specialisation carries it to the generic forms over the purely transcendental coefficient field $k(c_{i,a})$, so Conjecture 1 for $(n; d_1, \dots, d_r) = (4; 7, \dots, 7)$ and $6 \leq r \leq 21$ holds over such a field in every characteristic, and nothing is claimed over a finite field, where there are no generic forms in that sense. The certificate is a coprime pair: for each of ten endpoint cells (r, j) we exhibit two maximal square minors of one integer matrix whose determinants are coprime, so no prime divides both, so the rank is maximal modulo every prime at once; ten cells then spread over the whole interval by a characteristic-free propagation lemma, proved here. The family, the endpoint table and the propagation lemma are Wang–Zhang's, as is the characteristic-zero case; inside $6 \leq r \leq 21$ the characteristic-2 slice is already an immediate consequence of their printed rank table over $\mathbb{F}_2$ together with their propagation lemma, which carries no field hypothesis. What is added here is the coprime-pair certificate and, with it, the remaining primes.

1. THE CONJECTURE

Let k be a field, $S = k[x_1, \dots, x_n]$ with the standard grading, and let $f_1, \dots, f_r$ be forms of degrees $d_1, \dots, d_r$. Write $R = S/(f_1, \dots, f_r)$. Fröberg's conjecture predicts the Hilbert series of R when the f_i are *generic*, meaning that their coefficients are algebraically independent over the prime field of k . In the numbering of the Fröberg–Lundqvist survey [1, Conjecture 1] the statement reads, verbatim:

$$R(z) = \left(\prod_{i=1}^r (1 - z^{d_i}) / (1 - z)^n \right)_+, \text{ where } (\sum_{i \geq 0} a_i z^i)_+ = \sum_{i \geq 0} b_i z^i, \text{ where } b_i = a_i \text{ if } a_j \geq 0 \text{ for all } j \leq i \text{ and } b_i = 0 \text{ otherwise.}$$

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Conjecture 1 sits on p. 416 of [1]. The survey assumes $k = \mathbb{C}$ unless otherwise stated (p. 415); its Section 7, *Positive characteristic* (p. 426), is where it is otherwise stated, and the positive-characteristic cases recorded there are $n \leq 2$ together with $n = 3$ over an infinite field, followed by the sentence “We believe that Conjecture 1 is true for any field.” So the conjecture is posed over an arbitrary field by its own authors. For $n = 4$ and $d = 7$ in characteristic p , $r \leq 4$ is the monomial complete intersection $(x_1^7, \dots, x_4^7)$.

2. THE RESULT

Throughout, $n = 4$, $d = 7$, $S = k[x_1, x_2, x_3, x_4]$, and for forms $F_1, \dots, F_r \in S_7$ we write $I(r) = (F_1, \dots, F_r)$ and

$$\mu_{r,j}: (S_{j-7})^r \longrightarrow S_j, \quad (h_1, \dots, h_r) \longmapsto \sum_{i=1}^r h_i F_i,$$

whose image is $I(r)_j$, so that

$$(1) \quad \dim_k(S/I(r))_j = \binom{j+3}{3} - \text{rank } \mu_{r,j}.$$

For the equal-degree case at hand put

$$a_j(r) = [z^j] \frac{(1-z^7)^r}{(1-z)^4} = \sum_{i \geq 0} (-1)^i \binom{r}{i} \binom{j-7i+3}{3},$$

and let $q(r)$ be the largest j with $a_0(r), \dots, a_j(r)$ all positive. The truncation in Conjecture 1 then says: the conjectured series has coefficient $a_j(r)$ for $j \leq q(r)$ and 0 for $j > q(r)$. (If some $a_j(r)$ vanishes, the two readings of “ $a_j \geq 0$ ” in the conjecture’s truncation rule agree, because the coefficient printed is 0 either way; for $6 \leq r \leq 21$ no coefficient $a_j(r)$ with $j \leq q(r)$ vanishes, and this is re-derived in the verification.)

Theorem 1. *Let k be any field and $S = k[x_1, x_2, x_3, x_4]$. Let $G_1, \dots, G_{21} \in \mathbb{Z}[x_1, x_2, x_3, x_4]_7$ be the 21 septics printed in Appendix A, each a sum of distinct degree-7 monomials with coefficient 1, and read them in S by reducing their coefficients modulo the characteristic of k . Then for every r with $6 \leq r \leq 21$ the quotient $S/(G_1, \dots, G_r)$ has Hilbert series*

$$\sum_{j=0}^{q(r)} a_j(r) z^j = \left(\frac{(1-z^7)^r}{(1-z)^4} \right)_+,$$

the series predicted by Conjecture 1 for $(n; d_1, \dots, d_r) = (4; 7, \dots, 7)$.

Theorem 1 is a statement about one explicit integral tuple; no genericity enters it. Genericity enters only through the specialisation argument below, which is the mechanism behind the criterion that [1, p. 416] states for this purpose: “if one finds an algebra with the conjectured series for some $(n, r, d_1, \dots, d_r)$, then the conjecture is proved for these values”.

Proposition 2. *Let k be a field, $K = k(c_{i,a})$ the purely transcendental extension of k generated by indeterminates $c_{i,a}$ indexed by $1 \leq i \leq r$ and the monomials x^a of degree 7, and $F_i = \sum_a c_{i,a} x^a \in K[x_1, \dots, x_4]_7$ the generic forms. If some specialisation of the $c_{i,a}$ to elements of k makes $\text{rank } \mu_{r,j}$ equal to $\min(\dim S_j, r \dim S_{j-7})$ for every j , then the same holds for the generic F_i over K , and hence $K[x]/(F_1, \dots, F_r)$ has the series of Theorem 1.*

Proof. The entries of the matrix of $\mu_{r,j}$ are $\mathbb{Z}$ -linear in the $c_{i,a}$. Let $s = \min(\dim S_j, r \dim S_{j-7})$. By hypothesis some $s \times s$ minor is nonzero after the specialisation; that minor is a polynomial in the $c_{i,a}$ with coefficients in the prime field, and it does not vanish at a point of k^N , hence it is not the zero polynomial, hence it is nonzero in K . So the generic rank is at least s , and it is at most s trivially. Now (1) computes the generic Hilbert function. $\square$

Theorem 1 supplies the hypothesis of Proposition 2 over a field of any characteristic: for the specialisation sending $c_{i,a}$ to the coefficients of G_i , its dimension formula read through (1) says exactly that $\text{rank } \mu_{r,j}$ is the smaller of $\dim S_j$ and $r \dim S_{j-7}$ in every degree j . So for every field k Conjecture 1 holds for $n = 4$, $d_1 = \dots = d_r = 7$ and $6 \leq r \leq 21$ over the coefficient field $K = k(c_{i,a})$, whose characteristic is that of k and hence arbitrary. Over a *finite* field there are no generic forms in the above sense, and we claim nothing about Conjecture 1 there; what Theorem 1 gives over $\mathbb{F}_p$ is the explicit tuple itself.

3. THE WITNESS FAMILY

Order the $\binom{10}{3} = 120$ monomials of degree 7 in x_1, x_2, x_3, x_4 by *decreasing lexicographic order of their exponent vectors*, so that the list begins $[7, 0, 0, 0]$, $[6, 1, 0, 0]$, $[6, 0, 1, 0]$, $[6, 0, 0, 1]$, $[5, 2, 0, 0]$, $\dots$ and ends $[0, 0, 1, 6]$, $[0, 0, 0, 7]$; call this ordered list MON_7 and index it from 0. Each form of the family is a sum of distinct monomials from MON_7 , all coefficients 1, and is written as a 30-digit hexadecimal *mask*: the monomial with index i occurs in the form if and only if bit i — the coefficient of 2^i — of the mask is 1. The family has 120 members; Appendix A prints the masks of the 21 used in Theorem 1, whose support sizes run from 49 to 78, and all 120 masks are listed in the accompanying program `verify.py`.

To make the convention checkable by hand, here is G_1 in full. Its mask is 49ed660e53a564e3fb9f22d51e14f7 and it is the sum of x^a over these 65 exponent vectors a :

```
[7,0,0,0] [6,1,0,0] [6,0,1,0] [5,2,0,0] [5,1,1,0] [5,1,0,1]
[5,0,2,0] [4,3,0,0] [4,2,0,1] [4,0,2,1] [4,0,1,2] [4,0,0,3]
[3,4,0,0] [3,2,1,1] [3,1,3,0] [3,1,1,2] [3,0,4,0] [3,0,3,1]
[3,0,1,3] [2,4,0,1] [2,3,0,2] [2,2,3,0] [2,2,2,1] [2,2,1,2]
[2,2,0,3] [2,1,2,2] [2,1,1,3] [2,1,0,4] [2,0,4,1] [2,0,3,2]
[2,0,2,3] [2,0,1,4] [2,0,0,5] [1,6,0,0] [1,5,1,0] [1,4,0,2]
[1,3,3,0] [1,3,2,1] [1,2,4,0] [1,2,1,3] [1,2,0,4] [1,1,4,1]
[1,1,2,3] [1,0,6,0] [1,0,4,2] [1,0,3,3] [1,0,2,4] [0,7,0,0]
```

[0,6,0,1] [0,5,0,2] [0,4,3,0] [0,4,2,1] [0,3,1,3] [0,3,0,4]
 [0,2,3,2] [0,2,2,3] [0,2,0,5] [0,1,5,1] [0,1,4,2] [0,1,2,4]
 [0,1,1,5] [0,1,0,6] [0,0,7,0] [0,0,4,3] [0,0,1,6]

The witness for a given r is the *prefix* $G_1, \dots, G_r$. Writing each form as the sorted list of its exponent vectors and the family of 120 as the list of those lists, the canonical JSON encoding $[[[7,0,0,0], [6,1,0,0], \dots]], \dots]$ (no whitespace) has

`sha256 = f93339e7b30deb19b81fe74fa3b79a7859cf8da9a4c6fadbd740f98c975c8e20.`

This is the digest of the `forms` array of the ancillary certificate published with [2]: the family is theirs.

4. TEN CELLS, AND WHY THEY SUFFICE

Lemma 3 (endpoint propagation; [2]). *Let k be any field, $S = k[x_1, \dots, x_n]$, and $G_1, \dots, G_b \in S_d$. Let $1 \leq a \leq b$ and $q \geq d$, and suppose*

- (i) $\mu_{b,q}$ is injective, and
- (ii) $\mu_{a,q+1}$ is surjective.

Then for every r with $a \leq r \leq b$: $\mu_{r,j}$ is injective for $d \leq j \leq q$, and $(S/I(r))_j = 0$ for every $j \geq q+1$. Consequently

$$\dim_k(S/I(r))_j = \begin{cases} \binom{j+n-1}{n-1}, & 0 \leq j < d, \\ \binom{j+n-1}{n-1} - r \binom{j-d+n-1}{n-1}, & d \leq j \leq q, \\ 0, & j > q. \end{cases}$$

Proof. (a) For $r \leq b$, $\mu_{r,q}$ is the restriction of $\mu_{b,q}$ to the subspace $\{h_{r+1} = \dots = h_b = 0\}$, so it is injective by (i). (b) Let $d \leq j \leq q$ and let $h = (h_1, \dots, h_r)$ satisfy $\mu_{r,j}(h) = 0$ with $h \neq 0$. Choose any monomial m of degree $q-j$. Then $mh \neq 0$, because S is a domain, and $\mu_{r,q}(mh) = m \mu_{r,j}(h) = 0$, contradicting (a). So $\mu_{r,j}$ is injective. (c) For $r \geq a$ the image of $\mu_{r,q+1}$ contains the image of $\mu_{a,q+1}$, which is all of S_{q+1} by (ii); hence $I(r)_{q+1} = S_{q+1}$. (d) For $j \geq q+1$, $S_j = S_{j-q-1} \cdot S_{q+1} = S_{j-q-1} \cdot I(r)_{q+1} \subseteq I(r)_j$, so $(S/I(r))_j = 0$. The displayed dimensions are then (1) with rank $\mu_{r,j} = r \dim S_{j-d}$ in the injective range. No hypothesis on k is used beyond S being a domain, which holds over every field; in particular the lemma transports to characteristic p unchanged. $\square$

For $6 \leq r \leq 21$ the value $q(r)$ is constant on five blocks:

block of r	{6}	{7, 8}	{9, 10}	{11, ..., 14}	{15, ..., 21}
$q(r)$	13	12	11	10	9

so Lemma 3 needs one injective endpoint (b, q) and one surjective endpoint $(a, q+1)$ per block: ten cells in all, listed in Table 1. The blocks cover $\{6, \dots, 21\}$ with no gap and no overlap, and the arithmetic behind the table — including that no second Koszul term $\binom{r}{2} \binom{j-14+3}{3}$ enters $a_j(r)$ in the ranges used, the one exception being the surjective endpoint $(6, 14)$, where $a_{14}(6) =$

$680 - 720 + 15 = -25 < 0$ so that the truncation still begins at $q + 1$ — is re-derived in the verification.

TABLE 1. The ten endpoint cells. In each row the matrix is that of $\mu_{r,j}$ in the monomial bases, of size $\dim S_j \times r \dim S_{j-7}$; $s = \min$ of the two, so “injective” means $\text{rank} = s = \# \text{columns}$ and “surjective” means $\text{rank} = s = \# \text{rows}$. The last two columns give the bit lengths of the two exhibited minor determinants and their gcd.

r	j	matrix	s	endpoint	$\log_2$ of the two $ \det $	gcd
6	13	560×504	504	injective	800, 789	1
6	14	680×720	680	surjective	1070, 1073	1
8	12	455×448	448	injective	745, 737	1
7	13	560×588	560	surjective	905, 912	1
10	11	364×350	350	injective	592, 570	1
9	12	455×504	455	surjective	756, 752	1
14	10	286×280	280	injective	475, 467	1
11	11	364×385	364	surjective	611, 613	1
21	9	220×210	210	injective	365, 362	1
15	10	286×300	286	surjective	484, 488	1

Lemma 4. *Let M be an $R \times C$ integer matrix and $s = \min(R, C)$. Suppose two $s \times s$ submatrices N_1, N_2 of M satisfy $\gcd(\det N_1, \det N_2) = 1$. Then for every field k the image of M in $k^{R \times C}$ has rank exactly s .*

Proof. If $\text{char } k = p > 0$ then p cannot divide both determinants, so one of them is a unit in $\mathbb{F}_p \subseteq k$; if $\text{char } k = 0$ then $\gcd(\det N_1, \det N_2) = 1$ forces $\det N_1 \neq 0$, and a nonzero integer is nonzero in k . Either way some $s \times s$ minor of the image of M is nonzero, so its rank is at least s , and it is at most s by size. $\square$

Lemma 4 is the whole of the new input. A single maximal minor that is nonzero modulo one prime — for instance nonzero modulo 2, which is what the rank tables of [2] record — certifies that the integer determinant is nonzero, hence maximal rank over $\mathbb{Q}$ and over every field of characteristic zero, and *nothing* at any other prime. Two maximal minors of the same integer matrix with coprime determinants certify maximal rank at every prime at once.

Proof of Theorem 1. For each of the ten cells (r, j) of Table 1 the accompanying program `verify.py` holds two index sets, each naming an $s \times s$ submatrix of the matrix of $\mu_{r,j}$ built from the prefix $G_1, \dots, G_r$ over $\mathbb{Z}$, together with the two integer determinants, whose bit lengths and gcd are the last two columns of Table 1. Rows there are indexed from 0 by MON_j in decreasing lexicographic order, and columns from 0 so that column $i \cdot \dim S_{j-7} + m$

carries the m -th monomial of MON_{j-7} multiplying G_{i+1} , with $0 \leq i < r$; at an injective endpoint the minor uses every column and the rows outside the listed set, at a surjective endpoint every row and the columns outside it. The two determinants are coprime in every one of the ten cells. By Lemma 4 the rank of $\mu_{r,j}$ is s over every field k , which is injectivity at the five cells with $s = \# \text{columns}$ and surjectivity at the five cells with $s = \# \text{rows}$. Feeding the two endpoints of each block into Lemma 3 gives, for every r in that block and every field k , the dimensions displayed there; and those dimensions are $a_j(r)$ for $j \leq q(r)$ and 0 beyond, which is the truncated series. $\square$

5. SCOPE, AND WHAT IS TAKEN FROM ELSEWHERE

The interval. For $r = 21$ one has $a_9(21) = 10 > 0$ and $a_{10}(21) = -134$, so the predicted series has support through degree 9 and vanishes from degree 10 on; establishing the conjecture at $r = 21$ accordingly requires surjectivity in degree 10, which is what the surjective endpoint (15, 10) of Table 1 supplies through Lemma 3. For $r = 22$ one has $a_9(22) = 0$, so the predicted series stops at degree 8. Nothing is claimed here for $r \geq 22$, and nothing for $r = 5$; at the lower end $r \leq 4$ is the monomial complete intersection.

What is taken from [2], and what is added. The family of Appendix A is theirs, the block table is theirs, and Lemma 3 is theirs; the second is re-derived here from the series arithmetic, and the third is proved above rather than cited so that the argument is self-contained. Their Theorem 1.1 settles the characteristic-zero case, and their Table 3 records maximal rank over $\mathbb{F}_2$ for the ten cells of Table 1. Since their propagation lemma is stated with no hypothesis on the field, the characteristic-2 slice of Theorem 1 for $6 \leq r \leq 21$ is an immediate consequence of that table together with that lemma; they do not assert it, their Section 5 claiming characteristic zero only. What is added here is Lemma 4 and the coprime pairs, which cover the remaining primes, and the fact that the witness is a single explicit tuple over $\mathbb{F}_p$ rather than an open set of tuples. The integer determinants were computed independently of [2], which reports ranks modulo 2 and no integer determinants.

6. VERIFICATION

The folder that accompanies this note contains one program, `verify.py` (Python 3.9 or later, standard library only, no third-party package and no external data file), and a recorded run of it, `verify.output.txt`. The program holds the 120 masks, the ten cells of Table 1 and the twenty index sets and twenty integer determinants, and re-derives every quantity claimed here: it decodes the masks and recomputes the `sha256` of §3; re-derives $q(r)$ for $r = 6, \dots, 21$, the five blocks, and the claim that the ten cells of Table 1 are exactly the ten endpoint cells; rebuilds each of the ten integer matrices from the decoded forms, checks its shape against $\dim S_j$ and

$r \dim S_{j-7}$, extracts the twenty named minors and recomputes their determinants from scratch; takes the ten gcds; and, independently of the gcd argument, recomputes the rank of each of the ten matrices over $\mathbb{F}_2$, $\mathbb{F}_3$ and $\mathbb{F}_5$. The arithmetic is exact: modular Gaussian elimination over 62-bit prime fields and Chinese remaindering against the integer Hadamard bound $\det(N)^2 \leq \prod_i (i\text{-th row sum of } N)$, valid because every entry is 0 or 1, so that the modulus used exceeds $2|\det N|$ and the lift is the true integer; then one Euclid gcd per cell. No floating point enters any decision. The recorded run reports

VERDICT: ALL 222 CHECKS PASS

and the program exits 0 if and only if every check passes.

Two limits are recorded in the program's own output. It reads no file and no network, so it does not fetch the ancillary file of [2]: the `sha256` of §3 is recomputed from the masks it holds, and its equality with the published digest is a claim a referee must check against arXiv. And it checks nothing about $r \leq 5$, $r \geq 22$, or any (n, d) other than $(4, 7)$.

APPENDIX A. THE TWENTY-ONE FORMS

The 21 forms used in Theorem 1, as 30-digit hexadecimal masks over the ordered basis MON_7 of §3. Bit i of the mask is 1 if and only if the i -th monomial of MON_7 occurs in the form.

```
G1 49ed660e53a564e3fb9f22d51e14f7
G2 2728e9577f9976d66df364ad739e9b
G3 e4642c6e3d0c00c837aad525608035
G4 a697391e0adb52a5bbdb4606db8a6
G5 476783c09aadb8c07d24678ea4bbdd
G6 b29ddfdd727b6b736069bafd78deff
G7 53c2fe37a5bc1a5e7c15ddfecc5a3
G8 8bc155a7fb2fdaca14f9b5219eeaf4
G9 6703cbffcc1ee6058f28ea9114ed06c
G10 4c1f42a4ceb4857b5f07e66cedd8ef
G11 4cc658bc60373a0e6d46cc4493e1c5
G12 f380fa591a18fe94d6e091acd9979a
G13 d654523b92b9bb4034ce8b4d293176
G14 44e3a772aea91b2853a289bd834cc9
G15 2b2671f51fac829ce9e991ce47bc38
G16 b53b6f0b4b896b8cbf7f55fb902070
G17 4495e190e5533494c2071c137aa59d
G18 73837221ac7cb93029c29f2e30e086
G19 26ed68177571506497c38afb7b43fd
G20 42be9cc2a6dbe2f55faf1411447aa5
G21 9fe550885e47ed36507d97205539e0
```

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- [1] R. Fröberg and S. Lundqvist, *Questions and conjectures on extremal Hilbert series*, Rev. Un. Mat. Argentina **59** (2018), no. 2, 415–429. doi:10.33044/revuma.v59n2a10. All quotations above are from this text.

- [2] Y. Wang and L. Zhang, *Fröberg's conjecture for quintics and septs in four variables*, arXiv:2608.24797. The origin of the family of Appendix A, of the block table, and of Lemma 3. A preprint, not refereed.